INTERNSHIP/CAPSTONE PROJECT IN SOFTWARE ENGINEERING

BRIEFING ON PROJECT REQUIREMENTS FOR SE33

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Agenda

- Objectives
  - To brief the participants of MTech SE on the requirements, conduct and assessment of the Internship/Capstone project
- Topics
  - Objectives of Internship/Capstone Project
  - Requirements of Internship/Capstone Project
  - Approach for Project Conduct
    - Project Milestones
    - Progress Report for Internship/Capstone Project
  - Selection of Internship/Capstone Project
    - Proposal for Internship/Capstone Project
    - Typical Examples of Internship/Capstone Project
  - Assessment of Internship/Capstone Project

Objectives of Internship/Capstone Project

- Enable students to bring together all the topics they have studied and apply the **tools, techniques and methods** to **architect and build** smart systems, products and platforms that are secure and scalable
  - **Added value** in some aspects like Agentic AI, LLM, machine learning, analytics, safety-criticality, real-timeliness, fault tolerance, hardware integration, robotic and vision, etc. is expected
  - Act as a mechanism to enable **students** to **demonstrate their mastery** in the subject matter and their ability to be an **efficient practitioner**
- Allow **sponsoring organizations** the opportunity to **experience and assess** new technologies, techniques and methods
- Provide the final component of the MTech Software Engineering degree

Requirements of Internship/Capstone Project

- Project spans areas covered by a few **certificates** with some **added value**
- Scope of work focuses on the **architecting** of the solution as well as the **building of a significant slice** of the functionality
  - Architecture should cater to a few releases
  - Constructed slice of the software could target **MVP** first

Batch	**Part-time**	**Full-time**
Duration	8 months	Approx. 5 months (20 weeks)
Schedule	16 Mar 2026 ~ 30 Oct 2026 *If desirable and feasible, the team can always start informally earlier. However, it should not end earlier.*	23 Mar 2026 – 14 Aug 2026
Mode	Off-site project	On-site project

Approach for Project Conduct

- **3-phased** approach
- **Incremental** and **iterative** with **agile** practices
- A phase can be divided into **multiple Sprints**. The Sprint duration is decided by the student team and/or the company sponsor

Phase	Typical Part-time Schedule	Typical Tasks
0	4 weeks with 1 sprint	Product/Platform Roadmap, Solution Architecture and Product/Platform Backlog
1	14 weeks with a few sprints	Sprint Planning, Sprint Conduct, Software Design, DevOps Pipeline Design, Testing, Sprint Review and Sprint Retrospective
2	14 weeks with a few sprints	Same as above

Internship Milestones (For Full-Time Students)

Full-time Date	Agenda	Description
25 Sep 2025	Project Briefing	The participants are briefed on the requirements, conduct and assessment of the capstone/internship project
31 Jan 2026	Proposal Submission	The participants submit their proposals for review by the lecturers
15 Feb 2026	Proposal Review	The lecturers provide the review comments. The participants revise the proposals, if necessary, and kick start their projects
23 Mar 2026 ~ 14 Aug 2026	Project Conduct	The participants conduct their projects and consult the lecturers on issues. Each group submits a fortnightly progress report for review by the lecturers
25 May – 29 May 2026 (Tentative)	First Project Presentation	The participants present their projects after Phase 1
7 Jun 2026	First Submission	The participants should submit their presentation deck, initial draft report, demo videos and code artifacts
17 Aug – 21 Aug 2026 (Tentative)	Final Project Presentation	The participants present their projects after Phase 2
31 Aug 2026	Final Submission	The participants submit their presentation deck, project report, incorporating remarks gathered during their project presentations, demo videos and code artifacts

Capstone Milestones (For Part-Time Students)

Part-time Date	Agenda	Description
25 Sep 2025	Project Briefing	The participants are briefed on the requirements, conduct and assessment of the capstone/internship project
31 Jan 2026	Proposal Submission	The participants submit their proposals for review by the lecturers
15 Feb 2026	Proposal Review	The lecturers provide the review comments. The participants revise the proposals, if necessary, and kick start their projects
16 Mar 2026 ~ 30 Oct 2026	Project Conduct	The participants conduct their projects and consult the lecturers on issues. Each group submits a fortnightly progress report for review by the lecturers
27 Jun 2026 (Tentative)	First Project Presentation	The participants present their projects after Phase 1
11 Jul 2026	First Submission	The participants should submit their presentation deck, initial draft report, demo videos and code artifacts
10 Oct 2026 (Tentative)	Final Project Presentation	The participants present their projects after Phase 2
30 Oct 2026	Final Submission	The participants submit their presentation deck, project report, incorporating remarks gathered during their project presentations, demo videos and code artifacts

Detailed Project Agenda

Phase	Deliverables	Typical Tasks
0	• Product/Platform Roadmap (including business objectives and strategy, features/services and scope) • Product/Platform Backlog (including user stories or use cases and architectural constraints) • Solution Architecture (initial) (including architectural risks and mitigation)	• Determine the roadmap of the product/platform • Elicit and document user stories and architectural constraints of the product/platform • Craft the initial solution architecture • Optionally conduct feasibility study and prototyping
1	• Sprint Plan • Spring Backlog • Burndown Chart • Sprint Review • Sprint Retrospective • Software Design (for critical user stories) • DevOps Pipeline Design • Test Cases & Results • Codebase • Product/Platform Roadmap (updated) • Solution Architecture (updated) • First Presentation • Peer Assessments	• Conduct regular sprint activities (review product backlog, perform sprint planning, produce the sprint backlog, conduct daily stand-up meeting, update burndown chart daily, conduct sprint review and conduct sprint retrospective) • Design the critical user stories • Design the DevOps pipeline • Construct the solution for the user stories • Create test cases for user stories and test them against the solution • Update the product/platform roadmap • Update the solution architecture
2	Same as above	Same as above

Progress Report for Internship/Capstone Project

- Submit to Canvas periodically for review by NUS-ISS
  - Full-time: **Fortnightly**; the first progress report is due on **7 Apr 2026**
  - Part-time: **Monthly**; the first progress report is due on **1 May 2026**
- The report should include the following content

Section	Content
Project Title	Title
Date	Date of report
Summary of Work Done	Sprint Backlog (List of tasks for the Sprint conducted)
Effort Expended	Hours expended by each member since last report
Problems Encountered	Issues that might need assistance from ISS lecturers
Plan for the Next Reporting Duration	List of tasks planned for the next report duration (one month for part-time and fortnight for full-time)

Selection of Internship/Capstone Project

- Participants are to form a group of **4-5 members** by signing up to a *Class Groups* in Canvas
  - The Canvas module may only be ready and made available for any Internship/Capstone matters in early 2026
  - It is generally helpful if a team comprises participants that had taken various specialist certificates
- **Participants** are encouraged to **source** for suitable projects, either from within your **own organisations**, your **own ideas** or from **other approved sources**
- **NUS-ISS** would also be **soliciting** projects from its contacts. Suitable proposals would be **circulated** amongst all the students. Students are to discuss further with the sponsors in team
- **NUS-ISS** will provide **guidance** by reviewing the suitability of a project

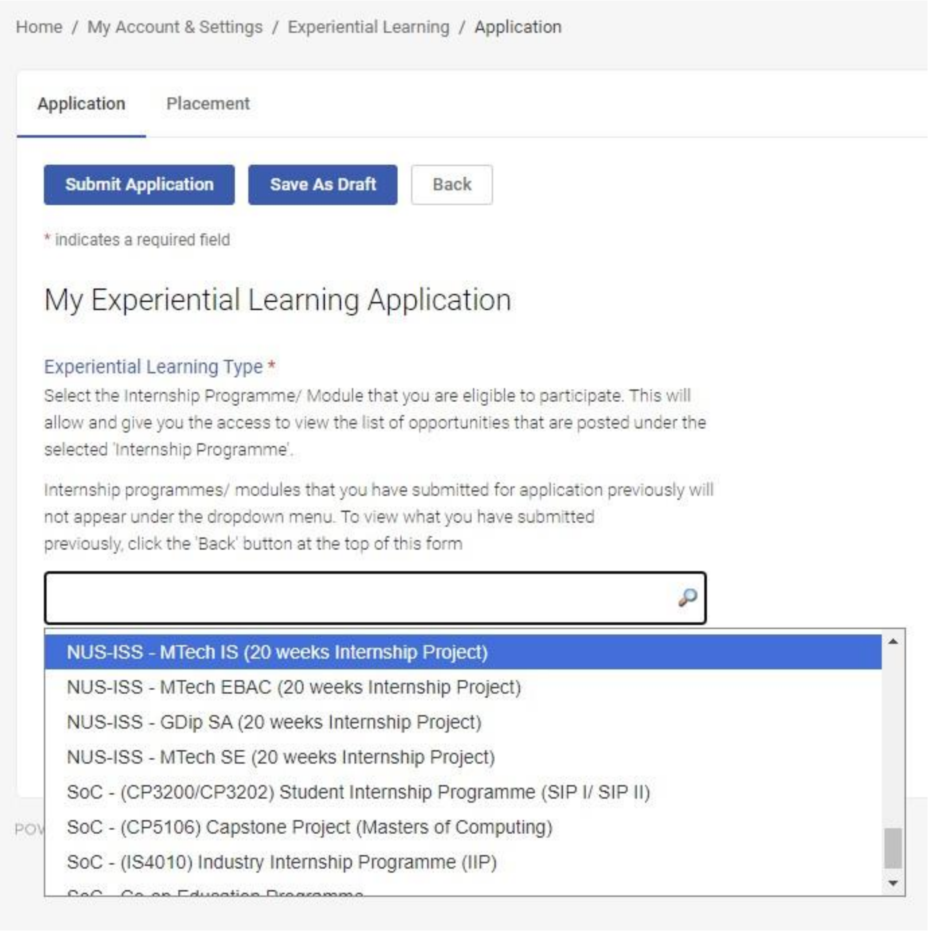
TalentConnect Key Steps Quick Guide – NUS-ISS

- Step 1: Login to TalentConnect (https://nus-csm.symplicity.com/students)

- Step 2: Go to **My Experiential Learning Record**

- Step 3: At 'Experiential Learning Type' field, select the NUS-ISS programmes that you would need to participate.

**After this step, you will be able to view the list of opportunities that are posted for the selected NUS-ISS programme.*



Proposal for Internship/Capstone Project

- Submit the proposal to Canvas by the proposal submission deadline for evaluation by NUS-ISS
    - Earlier submission and approval allows earlier conduct of the project
- The proposal should include the following content
    - A proposal template document is provided

Section	**Content**
Project Title	Title
Project Members	Names
Project Sponsor	Name, title, contact
Aims/Objectives	Describe the context and the business problem solved by the solution
Requirements Overview	Describe the requirements of the solution. It should be sufficiently detailed so that the breadth and complexity of the solution can be gauged. Value-add aspects should be highlighted. A write-up of 1-2 pages is typical
Resource Requirements	Highlight if the solution is constrained to specific hardware and software requirements
Methods and Standards	Highlight if the development process is constrained to specific engineering methods and standards

Typical Examples of Internship/Capstone Project–1

Problem description:

A leading **manufacturing** company wishes to develop a platform to manage their operations. They wish to build a **platform** where their **suppliers and logistics partners** can add value to their platform by providing online services. The manufacturing company also wishes to create a platform ecosystem to facilitate this. This requires development of useful **APIs** at the right granularity.

Deliverables and success criteria:

Develop a platform with reusable intelligent components and services. Development of APIs useful for the partners to build the ecosystem. Consuming useful APIs from other pies like Banks etc. Successfully deploying the solution. Use of state of the art technology and frameworks such as Cloud, Microservices etc.

Typical Examples of Internship/Capstone Project–2

<u>Problem description:</u>

A mature **tech start-up** company wishes to **productize** its successful **security software**. It is in the process of developing their digital product strategy and product roadmap. The existing product needs to be **re-engineered** to make it maintainable, extensible and scalable. Robust **security controls** have to be incorporated in the product.

<u>Deliverables and success criteria:</u>

Impact analysis needs to be performed on the existing system to understand the re-engineering efforts. The existing system should be re-architected and re-engineered according to the needs of the product strategy and roadmap. Appropriate security controls needs to be identified and incorporated in the existing system. The software system must exhibit good software engineering practices required to make it an extensible and evolving product.

Typical Examples of Internship/Capstone Project–3

Problem description:

A **transport operator** wishes to undertake building a **smart system** capable of **optimising the deployment of public transports**. It requires **monitoring the traffic conditions** and **predicting the number of vehicles** to be deployed at different times of the day in different regions of the city. The company wants to make use of the **sensor data** available from the **Lamp posts** and the **data collected by vehicles** plying in the regions. The system should be **scalable** to be deployed in large and small cities. The system should also be **extensible** for adding features such as providing specialised vehicles like ambulance, fire engines etc.

Deliverables and success criteria:

A well architected smart system capable of making use of the sensor data from the Lamp Post and other vehicles to yield optimised deployment of vehicles. The system should have used appropriate smart libraries and/or APIs to make sense of the data. The system should be able to give real-time responses and analysis. An intuitive and user friendly interface for interacting with the smart system is expected.

Typical Examples of Internship/Capstone Project–4.

Problem description:

A **big bank** wishes to engineer its **multi-channel data streams** for **analytics purposes** to improving their business and stay competitive. The bank wishes to analyse the **volumes of data** that are collected in **real time** from mobile devices, social media feeds and other channels. The data collected are a mix of **structured and unstructured**. A system that can engineer Data Lakes needs to be built.

Deliverables and success criteria:

Make real-time data streams useable by engineering Big Data. Build Data pipelines using techniques for ingestion, transformation, insight gaining etc. must be employed. The engineered system must be demonstrated by performing suitable data visualisation and analytics.

Assessment of Internship/Capstone Project

Assessment Component	Weight
First Presentation	10%
Second Presentation	10%
Project Deliverables	45% (including project demonstration)
Client Feedback	15%
Peer Assessments	20%
Total	100%

- The presentation and project deliverables would be graded by a team of NUS-ISS lecturers
- The participant must attain a **minimum overall score of 50%** (e.g. **Grade C**) **to pass the Internship/Capstone project**
- The participant must attain a **minimum overall score of 60%** (i.e. Grade B-) across all the qualifying certificates and Internship/Capstone project to be awarded a **Master of Technology** degree

Proposal Review

- **Proposal submitted** will be reviewed within 2 weeks from the date stated in **Internship/Capstone Milestones**
- If the **proposal doesn't meet the requirements**, the team will **receive an email** on the outcome and justification to furnish more details on a **resubmission of the project proposal.**
- **Student should only make a verbal acceptance or sign a contractual agreement with the company only if the project is accepted by the internship manager.**
- **Do take note:**

Once the capstone/internship project had been accepted verbally or contractually bound with the HR of the internship/capstone company, students are not allowed to change capstone/internship company or change project for any reason or to their own discretion.

Internship/Capstone guidelines, rules and regulations

If students require internship letters, please approach the Career Services Team for the letter.

For MTech SE students, please approach Pfeiffer Chung (pfchung@nus.edu.sg).

Internship/Capstone guidelines, rules and regulations

Students are to work on a potential project that meets the MTech SE Capstone/Internship project requirements.

Examples of projects that will not meet MTech SE project requirements and will not be approved:

- **Research project that focuses on writing research papers and/or doing prototype that doesn’t covers SE requirements across a series of grad certs**
- **Purely tester/quality engineer roles**
- **Product and project management roles**
- **Change request, change management and enhancement projects**

Internship/Capstone guidelines, rules and regulations

Once the capstone/internship project had been accepted verbally or contractually bound with the HR of the internship/capstone company, students are not allowed to change capstone/internship company or change project for any reason or to their own discretion.

Students who change their internship/capstone after they have accepted the internship/capstone will be subjected to report all violations to the Board of Examiners and the students may subject to receive warning letters, grade penalties, extended conduct of project and termination from the programme.

Internship/Capstone guidelines, rules and regulations

Students are to take note of the presentation period.
There shall be no change of presentation schedule slot once the Presentation schedule is shared to the cohort on the date and timing of presentation.
Student should flag out to the company, so that your supervisor is aware and make the necessary arrangement for you to participate in your presentation.

Student(s) feeling unwell and not able to take the presentation, must produce a valid Medical Certificate for re-schedule. Failure to do so will be subjected to report all violations to the Board of Examiners and the students may subject to receive warning letters, grade penalties, extended conduct of project and termination from the programme.

Internship/Capstone guidelines, rules and regulations

Students are to work on the internship/capstone project for the period defined. There shall not be any early termination or request to end the project before the internship period is over.

Students who terminate or end their internship early with/without the internship manager's approval will be subjected to report all violations to the Board of Examiners and the students may subject to receive warning letters, grade penalties, extended conduct of project and termination from the programme.

Internship/Capstone guidelines, rules and regulations

Student(s) who are terminated by internship/capstone company (other than company closed-down) during the official internship period should notify NUS-ISS internship manager(s).

If the termination is due to the following (non-exhaustive):

- **Inappropriate conduct or disrespect to staff**
- **Attitude, behavioral and character related issues**
- **Unable to perform assigned tasks or projects**
- **Incompliance or breach of company policy or non-disclosure agreements**

Student(s) will be subjected to report all violations to the Board of Examiners and the students may subject to receive warning letters, grade penalties, extended conduct of project and termination from the programme.

Internship/Capstone guidelines, rules and regulations

Student(s) must ensure that company project sponsor contact number, email address, company address are provided correctly.

We will do random check on the genuineness of the project and its sponsor.

Most importantly, at the end of the capstone project/internship, we will definitely contact the project sponsor for project feedback and individual feedback and assessment. This will constitute 15% of your overall grades.

If we are not able to obtain feedback from the sponsor, the sponsor grading will be <u>ZERO</u>.

Internship/Capstone guidelines, rules and regulations

Student(s) must declare if there is any <u>conflict of interest</u> with the company and project they are working on the following.

- **Student(s) have vested interest or are part of the partnership/founder of the company.**
- **The project sponsor(s) are related to any of the project team members, e.g. friends or relatives of the student(s).**

Internship/Capstone guidelines, rules and regulations

Student(s) internship or capstone project should be a software engineering project that covers the guidelines described in the slide "Presentation Guidelines".

If the project doesn't meet the guidelines, students may likely score poorly in the capstone/internship module. This will likely lead to students unable to graduate seamlessly.

1st Presentation Guidelines

1. Project overview and introduction (1 min)
2. Overall scope using a use case diagram (1 min)
3. Project roadmap & key milestones (1 min)
4. Complete project backlog of the project
5. Overall SPRINT effort to date. (Overall estimate effort versus actual effort of every member till-date for all SPRINTs conducted).
6. Tech Stack (1min)
7. Architectural constraints and decisions (1 min)
8. Diagrams - Logical & Physical architecture (1 min)
9. Deployment diagram (1 min)
10. Microservice architecture highlighted or explained using Domain Driven Design (2min)
11. Software design (1min)
12. DB design and noSQL design and collection objects (1 min)
13. CICD pipeline and and CICD diagram (1min)
14. Live Demo of work app, CICD demo. (3-6 min)
    Alternatively, MVP video of working app and CICD, i.e. video (3-6 min)
15. Unit Testing, Integration Testing, End-to-end testing, stress testing (2 min)
16. Management concerns, issues and mitigations (2 min)
17. Technical concerns, issues and mitigations (2 min)
18. Security concerns, mitigations and solutions (2 min)
19. AOB

Additional Guidelines I

- A team should not simply claim that it has carried out the various activities without showing suitable evidence. It can do so briefly during the presentation and in more detail in the project report for one or more architecturally significant use cases.
- A team should not just provide an architecture overview diagram for its architecture work. It should provide suitable detail using diagrams showing perspectives on functional elements, packing of functional elements into subsystems, deployment of subsystems onto nodes, etc.
- A team should normally provide a use case diagram as an overview of the functional requirements of the system.
- A team should normally provide class and sequence diagrams at class/object level for the design of its significant/complex use cases. By “design”, we don’t just refer to the persistence aspect, we expect to see how the use cases are to be implemented using the various frameworks used in the FE, BE, communication, etc.
- Without providing suitable evidence during the presentation and in the project report, the assessors would not be able to give marks on the activities that a team claims to have carried out.
- When discussing the resolution of an architectural/design problem, the team should articulate the problem and solution with some detail (e.g., diagrams), not just in words at high level.
- Confidentiality’ is not a reason not to show evidence of the work that they have done.
- Be prepared for the presentation to show the meat highlight above and also from the project guidelines sent in earlier announcement.
- Apply the techniques and knowledge you have acquired in our MTech program in your capstone project.

Additional Guidelines II

- Once accepted internship, there should be no further changes/withdrawal. Failure to do so will be penalised in terms on grades.
- If there is anything unclear, it is your responsibility to consult with the lecturers, or company-in-charge.

First Submission

- On Post Phase 0+1 presentation, you may be advised on the necessary changes or updates to your design.
- There will be an initial submission after your Phase 0+1 presentation.
- The deliverables to be submitted may include:
    1. Presentation slide
    2. Demo videos (i.e. App, CICD, Testing, Security, etc.)
        1. Each video should keep to 4-5 minutes.
        2. Video resolution should be at least 800x600 or 1920x1080 for the lecturers/reviewers to clearly review the demo and work done.
    3. Phase 0+1 Report (There is no report template. Students have to generate a comprehensive report that clearly explain their project and show the relevant artifacts to defend their project. Students should clearly address the questions raised by the lecturers during the presentation.)
    4. Code Artifacts + Repo URL (Students have to request their supervisor to drop an email to the Internship manager if they are not able to submit their source code as proof of work and artifacts.)

First Submission

- For team project, each individual team member will also need to submit a peer assessment.

  The reason for you to submit your Phase 0+1 report is to ensure that you document and update your report as you progress along the way to ensure that your final report submitted at the end of the project is more substantiate and of quality.

Assessment Rubrics I

Area	Maximum Duration	Assessment
Management Assessment	5 minutes	1. Project justification a. Project goals defined b. Pain-points and reason for building and enhancement highlighted (What is it trying to solve? What benefits it brings?) 2. Project scoping a. Project journey map b. Scope: Features or use cases c. Product backlog (user stories or use cases) 3. Project conduct a. Agile practices using SCRUM b. Tracking tool (e.g., JIRA) c. Product backlog (all user stories) d. SPRINT backlogs • All user stories completed • SPRINT goals achieved • Burndown charts 4. Project effort a. Overall efforts (man-days) for each member b. Planned Hours versus Actual Hours expended 5. Management issues and mitigation a. Client/Sponsor management

Assessment Rubrics II

Area	Maximum Duration	Assessment
Architecture Assessment	10 minutes	1. Logical architecture a. Logical architectural decisions b. Logical architecture overview c. Domain Driven Design diagrams d. Logical deployment diagram 2. Physical architecture a. Physical architectural decisions b. Technology stack c. Physical architecture overview (with infrastructural and networking details) d. Physical deployment diagram 3. Security assessment (threats and mitigation)

Assessment Rubrics III

Area	Maximum Duration	Assessment
Other Technical Assessment	5 minutes	1. Software design a. Use case diagram b. Class & sequence diagrams for analysis and design of a critical use case, highlighting on: i. Transition from analysis to design ii. Use of design patterns c. Data schemas and models

Assessment Rubrics III

Area	Maximum Duration	Assessment
Other Technical Assessment	10 minutes	2. DevSecOps a. CI/CD i. Pipelines and tools used ii. Unit testing (result artifacts) iii. Integration testing (result artifacts) iv. Load and stress testing (result artifacts) v. SAST (tool and result artifacts) vi. DAST (tool and result artifacts) b. Container Management i. Building and saving images ii. Image security (e.g., Trivy) iii. Interact and inspect containers iv. Container logs c. Vulnerability Assessment i. Resolution and rescan results of SAST ii. Resolution and rescan results of DAST d. Compliance as Code i. Infrastructure-as-Code (tools and artifacts) ii. Version control audit trails (e.g., Git history) e. Specific regulatory framework (e.g., SOC 2, HIPPA, GDPR)

Assessment Rubrics IV

Area	Maximum Duration	Assessment
Presentation Assessment	10 minutes	1. Demonstration of live system (Max. 5 mins video) 2. Demonstration of DevSecOps (Max. 5 mins video)

Assessment Rubrics V

Area	Maximum Duration	Assessment
Added Value	5 minutes	1. Meeting the minimum requirements of MTech SE internship/capstone project 2. Accepted by the sponsor and preferably going live 3. Explored one or more technically advanced or innovative areas like: a. Gen AI b. Agentic AI c. Data analytics d. Machine learning e. Real-time processing f. etc.

Final Submission Deliverables

- For team project that have **more than 2 persons**, each individual team member will also need to **submit a peer assessment** via the **Peer Assessment System**.
- **One or two member team** will **not require** to submit peer assessment.

- Code Artifacts + Repo URL (Students have to request their supervisor to drop an email to the Internship manager if they are not able to submit their source code as proof of work and artifacts.)

Final Submission Deliverables

- Students are to create the following presentation videos:
    1. TeamXX- Management Assessment.mp4
    2. TeamXX- Architectural Assessment.mp4
    3. TeamXX- Technical Assessment - Software Design.mp4
    4. TeamXX- Technical Assessment - DevSecOps.mp4
    5. TeamXX- Value Added Assessment.mp4
    6. TeamXX- Presentation Assessment App Demo.mp4
    7. TeamXX- Presentation Assessment CICD Demo.mp4

- Refers to the **Assessment Rubrics Slides 33 - 38** on the Duration of the video.
- Video should be in **HD (1920x1080) resolution**.
- Student are to turn on their camera and recording should record with speaker view showing the face clearly visible.
- For team project with more than 1 member, every member must contribute to the creation of presentation videos.
- Students are to put all videos created into one Zip file named <TeamXX-Project Title>.zip and make submission to Canvas.

What's next after the Capstone briefing?

Registration for the Master of Technology (MTech) programme via NUS Graduate Admission System (GDA3)

- Please proceed to register for the MTech programme in NUS GDA3 by 20 January 2026
  https://gradapp.nus.edu.sg/portal/app_manage

- Please note the respective deadlines:-
  - Apply in GDA3 by 20 January (2359 hours) 2026
  - Accept offer in GDA3 by 1 February 2026 (2359 hours)
  - Complete NUS Onboarding Process between 5 – 10 February 2026
  - Complete RC1000A & SE1000A on Canvas before 20 February 2026 (using the NUS student ID)

- This process is mandatory, as the POCS team will need to enrol the Capstone course in your student record to input your grade. Please ensure all required steps are completed by the stated deadline.
- Please adhere strictly to the respective deadlines as requests received after the deadline will not be considered.
- You will be added to Canvas from 12 February 2026 onwards.
- Please refer to the invitation email as well as the subsequent offer email for details on the above deadlines.
- Subsequent communications related to student matters will be sent only to your NUS email address.
- Please confirm your application for the Capstone course by submitting the GDA3 portal.